Rohan Mishra

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Professional Experience

Officer Cadet (Technical Projects) **Indian Army – OTA Gaya**
Mar 2025 – Present — All India Rank 5

- Architected secure hospital management system using **Java Spring Boot**, React.js, and PostgreSQL serving 500+ military personnel across 3 hospitals; eliminated 15+ hours/week of manual paperwork through streamlined RBAC-based workflows and encrypted patient data handling.
- Developed RESTful APIs with JWT authentication and role-based access control, enabling secure data exchange between hospital modules while maintaining HIPAA-equivalent compliance standards for military healthcare.
- Engineered **RFID-enabled armoury inventory system** with real-time tracking capabilities, reducing equipment audit time by 80% (from 4 hours to 45 minutes) and preventing critical stockouts through automated threshold alerts.
- Executed comprehensive testing strategy using JUnit and Postman, achieving 85%+ code coverage; applied microservices design patterns and enhanced PostgreSQL queries, reducing dashboard load times by 35%.
- Led cross-functional coordination for technical rollouts in Agile environment while mentoring 200+ cadets on software engineering best practices during pre-commissioning training.

Software Development Engineer **Jio Platforms, Navi Mumbai**
Oct 2023 – Feb 2025

- Designed and deployed microservices in **Python FastAPI** for Jio's CloudXP enterprise platform handling **high-volume production traffic** (99.9% uptime); reduced P95 API latency from 180ms to 95ms through async processing, connection pooling, and Redis caching.
- Established event-driven architecture using **Kafka for real-time streaming** (300K+ messages/hour) and RabbitMQ for async task processing, improving system throughput by 40% while maintaining fault tolerance across distributed services.
- Automated **CI/CD pipelines** with Jenkins, Docker, and Kubernetes, reducing deployment cycles from 2 weeks to 4 hours; achieved zero production incidents across 50+ releases through automated rollback mechanisms and canary deployments.
- Enhanced PostgreSQL performance for enterprise resource management database (50GB+, 500K+ records) serving Jio's cloud infrastructure; reduced critical report generation time from 45s to 8s through strategic indexing, query refactoring, and materialized views.
- Delivered unit and integration testing (JUnit, PyTest) achieving 80%+ coverage; integrated OpenTelemetry for distributed tracing, reducing mean time to resolution (MTTR) from 2 hours to 25 minutes.

Projects

- **Loan Acceptance Platform** (2023): Built ML-powered loan approval system using React.js, Next.js, Spring Boot, and XGBoost classifier achieving 85% precision on 50K+ loan applications; implemented JWT-based authentication and serverless APIs via AWS Lambda, reducing manual review workload by 60% while maintaining sub-3% fraud rate.
- **E-Commerce Analytics Dashboard** (2022): Created real-time sales analytics platform with React.js, Firebase, and Chart.js processing 100K+ daily transactions; designed RESTful APIs with modular architecture and improved rendering performance, reducing dashboard load time from 4.2s to 1.1s through lazy loading and data aggregation.

Technical Skills

Languages:	Java, Python, JavaScript/TypeScript, SQL, C++
Backend & APIs:	Spring Boot 3.x, FastAPI, Django, REST API Design, Microservices Architecture
Frontend:	React.js, Next.js, TypeScript
Databases:	PostgreSQL, MySQL, MongoDB, Redis (caching)
Messaging & Streaming:	Apache Kafka, RabbitMQ, Event-Driven Architecture
DevOps & Cloud:	Docker, Kubernetes, Jenkins CI/CD, AWS (Lambda, EC2, S3), GCP
Testing & Observability:	JUnit, PyTest, Postman, OpenTelemetry
Security:	OAuth2, JWT, RBAC, Data Encryption

Education

B.E. in Electronics & Telecommunication Engineering **SIES Graduate School of Technology, Mumbai**
2019 – 2023 — CGPA: 8.41/10.0 (First Class with Distinction)
Research Publication: "Stock Market Prediction Using Machine Learning" – Published in JETIR (July 2023) — Achieved 82% prediction accuracy using ensemble methods; deployed model on AWS with REST API serving 1K+ predictions